

ASP3231 Assignment 3

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Question 1

1a.

Wien's Displacement Law relates the wavelength at which a blackbody is most radiant to the inverse of its temperature. It is defined under the assumption that the object is a perfectly radiating blackbody, which is a typical assumption for analysing most stars. It follows that

$$\lambda_{\max} = \frac{2.898 \cdot 10^{-3} \text{m} \cdot \text{K}}{T}. \quad (1)$$

On the $t=1.43$ days graph in Figure 1, the wavelength that is emitted at the highest flux (intensity) is approximately $1.2 \cdot 10^3$ angstroms, or $1.2 \cdot 10^{-7}$ metres. Hence, the temperature of the object 1.43 days after the explosion is $\approx 2.415 \cdot 10^4$ K.

1b.

1c.

1d.

1e.

Question 2

2a.

2b.

2c.

2d.